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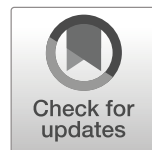
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# An effective approach for emotion detection in multimedia text data using sequence based convolutional neural network

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## Abstract

In the recent trends, the world has stepped into a multimedia era for enhancing business, recommendation systems, and information retrieval, etc. Multimedia data is highly rich in contents which express different human emotions. Several issues for emotion detection from multimedia images & videos have been addressed in this domain, but a very less effort has been applied for text data. The evaluation of deep learning has outperformed traditional techniques in sentiment analysis tasks. Inspired by the work done in the field of sentiment analysis, a deep learning based framework has been implemented on multimedia text data for the task of fine-grained emotion detection. The presented work introduces a new corpus which expresses different forms of emotions collected from a TV show's transcript. A manual annotation of the corpus has been conducted with the help of English expert annotators. As an emotion detection framework, this paper proposes a sequence-based convolutional neural network(CNN) with word embedding to detect the emotions. An attention mechanism is applied in the proposed model which allows CNN to focus on the words that have more effect on the classification or the part of the features that should be attended more. The main aim of the work is to develop a framework such a way to generalize to newly collected data and help business to understand the customer's mind and social media monitoring as it allows us to gain an overview of the wider public opinion behind certain topics. Experiments conducted on the dataset shows that the proposed framework correctly detects the emotions from the text with good precision and accuracy score.

**Keywords** Multimedia data · Emotion detection · Deep learning · Convolutional neural network · Deep neural network

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# 1 Introduction

In the current scenario of internet and web, a large amount of multimedia data is presented in an unprocessed manner. The multimedia data contains massive information stored in a specific manner like image, video, speech, and text files. Nowadays the task of identifying the patterns and making the prediction from the multimedia data is very crucial for business, recommendation and information retrieval systems. Several algorithms have been applied to extract significant data from multimedia files in order to perform similar searches, image classification, speech recognition, video emotion recognition, sentiment analysis, pattern recognition, etc. Textual data is one of the multimedia data presented over the internet. The data is in the form of customer reviews, political views, movie reviews, blogs, and social media posts. It contains lots of information which are needed to be extracted for different tasks like understanding the customer's view about a product, a voter's views about the political party, person's views about a movie, etc. Text data is highly used multimedia data for communicating and passing information from one source to another over the internet.

Moreover, communication is not only a way to express the informative contents; it also includes a speaker's attitude, sentiment, opinions, and hidden emotions. Emotions are in well-known described as subjective experience related to a person's mood, persona, nature, character, flair, state of mind, etc. They are the part of a person's daily life and are thoroughly expressed during his/her mode of communication; also, they provide surplus details about the content. The expressed emotions can be detected by facial expressions like fear, anger, and surprise, etc., the gestures made by him/her, in voices, the actions and also in the form of written text language.

The written text language is said to be an effective instrument to impart and pass on opinions and furthermore, a way to express emotions. The emotional state of mind of a writer is enclosed in the text which expresses his/her emotions directly through the words like happy, sad, joy or anger, etc. or indirectly through the sarcasm, slang, satire, and irony. Emotions are the critical component of human instinct and are generally being examined in the field of neuroscience, psychology, behavior science, health care, communication, computer science, cognitive science, human decision handling, etc. [79]. In computer linguistic, identifying emotions in text is relatively a new area of research which is rich in applications like tracking user's emotion about products or events [20], a breaking news [53], sports, election, political agenda or a bill passed in the parliament, to customer relationship management, question answering systems, health domain [37].

With the advent of ever-increasing and pervasive computational resources, their arises several huge opportunities for developing intelligent computational techniques to solve a variety of real-life problems [49, 92]. The emotion detection task attracted several researchers and motivated them to work in the field of developing intelligent computational techniques.

It aims at identifying and detecting the emotions from the text and in the end classifies them, in step with exclusive categories recommended by various researchers and analysts [16, 35, 68, 69, 85]. The past studies have aimed the problem for detecting emotion from multimedia data such as voice [67] and facial expressions through images [7] which is easy to be recognized but the problem occurs when the task deals with multimedia text data. This is because the exceedingly similar languages have often distinctive spelling and meaning also the way to express one's own views and opinion differs from person to person. For example, a sentence in a blog can have a fraction of happiness but with no presence of word happy or a word similar to happy. The advancement in technology and understanding of emotions are thereafter developing possibilities for computerized emotion recognition systems.

Emotion detection is fairly a challenging and hard hassle task in the field of natural language processing. It deals with the problems faced by the NLP components like entity recognition, word-sense disambiguation (WSD), co-reference resolution problem and anaphora resolution [8]. Another challenge is that the text rich in emotions can only be found on customer reviews on products, movies, political reviews, blogs, social networking sites like Facebook, Twitter, etc. and there is no surety that the text is enriched with necessary categories of emotions. Apart from the categories it also lacks punctuation, grammar and contains slang words. These challenges occur when a sentence level analysis is performed over short texts and tweets with limited characters. Another challenge is that it is not necessary that the text contains the exact word related to emotion, which is going to be detected, for example when text is sarcastic, or puns intended phrases. These phrases are ambiguous and sound similar but with different meanings, for example, *I nearly bought a clock today, but it wasn't the right time*. The biggest challenge faced in emotion detection task is the shortage of annotated emotion databases. For sentiment analysis where the task is to classify text as a positive, negative and neutral category, the large amount of annotated dataset exists. The existing database on sentiment analysis from twitter listed in [76] is SemEval 2013, Healthcare Reform, Obama-McCain Debate, Dialogue Earth-WA, etc. For emotion detection annotated dataset is very less in number, like ISER dataset [74], tale's corpus by [1], blog posts [4] and annotated headlines [80].

However, a better way to further investigate and explore the plausibility of text-based emotions, a more extensive dataset would be required. In this research work, a TV show transcript of a famous show Charmed has been collected as a training dataset. The transcript is the large corpus of textual data in the form of dialogues that contains a good amount of emotional content. They very well provide an insight related to the character's intentions and feelings. Moreover, they are written by the writer with the intention to inflame emotions, views, opinions portrayed by characters which make this type of textual content appropriate to be used in automatic emotion detection. The emotion models reported in different works of literature are presented in Table 1 [36]. Manual annotation of a new corpus has been conducted in this proposed work by utilizing the emotion model presented by the Ekman [16]. The help of English experts and students for the task of annotation has been taken for maintaining the quality of collected dataset. To classify the emotions from the dataset, a deep neural network; called sequence based convolutional neural network has been implemented in which the training data is passed in a sequential manner. This sequential manner helps the network to utilize the features of the previous sentence in order to detect emotion from the current sentence. In order to capture the semantic and context features of the input text the pre-trained word embeddings have been used as the first layer of the network which converts word into the vectors, where semantically similar words have same vector representation. The context features from the network are then passed to the attention model to discover whether the context is important or can be ignored. By implementing the attention mechanism and CNN, the model can pick the useful features from the sequences passed as the input. The results are compared with Long Short Term Memory (LSTM) and a random forest classifier on the basis of several evaluation metrics.

The organization of the rest of the paper is as follows: A literature works on emotion detection using machine learning and deep learning techniques are presented in Section 2. The corpus collection and building methodology, annotation task and inter-annotator agreement method have been presented in Section 3. The proposed framework of emotion detection and the experimental process has been presented in Section 4. The

**Table 1** Emotion categories or classes identified by authors [36]

| Muni [69]    | Tomkins [85] | Izard9]   | Plutchik [68] | Ekman [16] |
|--------------|--------------|-----------|---------------|------------|
| Anger        | Surprise     | Interest  | Anticipation  | Anger      |
| Energy       | Anguish      | Shyness   | Surprise      | Fear       |
| Disgust      | Interest     | Surprise  | Anger         | Sadness    |
| Astonishment | Fear         | Disgust   | Sorrow        | Disgust    |
| Love         | Disgust      | Sadness   | Acceptance    | Surprise   |
| Sorrow       | Shame        | Anger     | Joy           | Happiness  |
| Terror       | Anger        | Enjoyment | Fear          |            |
|              | Joy          | Shame     | Disgust       |            |
|              |              | Fear      |               |            |

results and outcomes are discussed in Section 5. Finally, precise conclusions and the scope of future work are mentioned in Section 6.

## 2 Related work

Emotion detection is firmly related to sentiment analysis and identifying the emotion categories from the text is a crucial task in NLP [55]. From the last few decades, a considerable amount of work has been done on sentiment analysis of text data but only limited work has been put on in the field of text emotion detection [36]. Several researchers have been attracted towards the study of emotions in the field of psychology, linguistics, social sciences and communication [77].

In this section, some of the past research work related to the text emotion detection is presented briefly. It has been seen that past research in emotion detection was mainly focused on finding the specific emotional keywords from the text. Yanaru et al. [86] applied a basic theory for subjective observation model and a fuzzy inference technique to an emotion processing system. They verified that the emotional words processed by the system actually resemble a real person and defined emotional categories corresponding to the keywords. A common sense affect model built by Liu et.al. [55] analyzed the emotive qualities of a text data and classified them into six emotion categories happy, angry, sad, fear, surprise, and disgust gave in Ekman's Model. For their approach, an expansively real-world common sense knowledge base was taken for the classification task of sentences in various emotion classes. Strapparava et al. [82] investigated the problem from the past study that there were not sufficient dimensions to identify the whole emotional concepts and lexical semantic approach doesn't allow distinguishing different senses of the same word. Therefore, they focused on extending the resources containing a set of effective concepts correlated with effective words making use of WORDNET relations. A corpus of 815,494 blog posts from Live journal was collected by Mishne, G [59] to classify different mood emotions [confused, curious, depressed, sick, excited, etc. using the SVM classification technique.

Read et.al.[25] discussed the annotation methodology for annotating book reviews as per the appraisal framework [87]. An automatic system based on Naïve Bayes classifier for detecting public emotions related to epidemics in multilingual unstructured data to understand the health-related information was developed by Jain et al. [38]. Several variations of latent semantic analysis were used for emotion analysis of news headlines extracted from news websites by Strapparava [81]. The hierarchical and flat classification approaches were applied

on blog corpus and children's stories dataset by Ghazi et al. [22] for the task of automatic classification of text into Ekman's six emotion classes. The multiple levels of the hierarchy were used by them, where at first, they classified them as positive and negative emotions and in the next level classified the emotion on a fine-grained level. The six emotion classes proposed by Ekman were categorized from Live journal blog dataset and news headline dataset. The emotions were tracked from emails by Mohammad and Yang [61] and they showed the usage of different emotion words in emails by males and females.

A Principle-based approach was proposed by Gao et al. [19] presented the experiments on emotion corpus of Chinese microblog (Weibo text) using a rule-based system. A machine learning based framework was proposed by Jain and Kumar [40] to classify emotions in multi-language data using different feature groups collected from sports, political, and healthcare dataset. The technique can improve the performance of different applications such as text summarization, question answering systems and categorization, etc. Jain and Kumar [36] also recognized six basic emotions of Ekman from customer reviews collected from online e-commerce websites like Flipkart, Snap deal and Amazon. In the last few years, Deep Neural Networks(DNN) [75] have revolutionized the field of natural language processing(NLP). The phrase "deep learning" is also used to describe these deep neural networks. Jain and Kumar [40] focused on investigating the merit of using deep models for sentiment analysis in customer reviews.

It has emerged as an effective and impressive machine learning technique where data features and representations are captured through multiple layers and produce a result which outperforms other states of the art techniques. The deep neural network architecture provides the deep understanding and learning capability of the features presented in data which in turn helps solves complicated AI task accurately and precisely.

## 2.1 Taxonomy of deep learning

Deep learning belongs to the area of an artificial neural network where the task is to learn the data using networks of multiple layers. In neural networks, a signal is processed in a similar way as in neurons of living beings [78]. It is extremely motivated by the structure of human cerebrum and comprises a substantial number of handling units called neurons. The network has an important feature of performing parallel processing of information by all links. The tremendous measure of inter-neural connections helps to fundamentally speed up the process of processing information. The connections between neurons are called synapses and each synapse has one option i.e. weight [14]. The weights can be adjusted based on inputs and outputs. In a neural network, neurons are organized in a layer and can figure out how to perform the task by altering the weights along the neurons. The neural network is divided into three layers, where, the first layer is the input layer for feeding data into the network. The hidden layer processes the output from the previous layer and features are extracted by applying certain operations on them [83]. The last layer is the output layer which classifies the data into predefined classes and displays the results. Each layer applies some linear and the non-linear activation function (sigmoid function, tanh function, ReLU function, etc.) to the previous layer to produce an output signal. Weights are multiplied with the number of input signals and the outcome is sent to the neuron. As the number of hidden layers in the neural network is increased it turns out to be more denser called deep neural network.

The Deep learning has been applied into several tasks like image classification [12], object detection [17] and semantic segmentation [23], machine translation [10], sentiment analysis [34] and speech recognition [30] and has shown the best performance for solving these tasks.

The deep learning network has been largely used for sentiment analysis tasks. CNN [51], Deep Belief Neural Networks [DBN] [29] and Recurrent Neural Networks[RNN] [24] are the special type of DNN architectures which are used in combination with each other or as a standalone which have shown the impressive performance for solving the various AI tasks. It makes a sense that the multiple hidden layers can be used effectively in order to detect the emotions from the data. As the human brain also detects the human emotions using the neurons and layers, so in this manner the DNN is most appropriate for taking care of this issue. The techniques for solving the text classification problem have been available in a large amount with the wide advancement of deep learning in this research area [91]. The most detailed study of CNN has been done among the different types of deep learning networks in this dimension.

## 2.2 Convolutional neural networks

Convolutional neural networks (CNN) have made deep learning one of the hottest topics of AI. CNN has been applied to a large number of tasks such as image classification, sentiment analysis, etc. because of its capability to detect the features using the deep hidden layers. Human visual cortex which is a type of visual mechanism inside the brain of animals has inspired the design of CNN. It was found by Hubel and Wisel [33] in 1959 for detecting the light in receptive fields the cells presented in the animal visual cortex are highly responsible substance. The CNN was designed to use on two-dimensional datasets like videos and images as it belongs to the multi-layered feed-forward neural network. It was originally employed in the field of computer vision proposed by Yann LeCun [52]. A multilayer ANN called LeNET5 was developed by them to classify the handwritten digits. Since past few years, CNN has been applied to classify and predict text data for task of the sentiment analysis, document classification, sentence classification, etc.

Kim [45] proposed CNN architecture and trained the architecture using pre-trained word vectors for solving various tasks of sentence classifications that include sentiment, subjectivity and question type classification. Their results show that for a better performance of CNN a well-tuned hyper-parameter can be adapted. Kalchbrenner et al. [44] proposed a dynamic CNN for the semantic modeling of sentences. A K max pooling operation over linear sequence was used by them in the architecture. The task of semantics modeling was performed on the question and twitter datasets. Ruder et al. [73] applied CNN architecture for aspect-based sentiment analysis by concatenating an aspect vector along with the word embeddings on the input dataset for achieving competitive results. A semi-supervised framework with CNN for test categorization was proposed by Johnson and Zhang [42] proposed. Their model achieved better results on sentiment classification and topic classification task as compared to previous models. Research by You Q et al. [90] was motivated through the need controlling of comprehensive social multimedia content and employ both textual and visual SA techniques for combined textual visual sentiment analysis. Ouyang et al. [65] proposed a framework called word2vec and CNN. First, they compute vector representation of words using word2vec and designed CNN architecture for sentiment analysis task. The framework was verified on the dataset from [rottentomatoes.com](http://rotentomatoes.com) which contains movie review corpus.

The CNN consists of few layers such as a convolutional layer for feature extraction, pooling layer for reducing the spatial size and a fully connected layer where the features from data are extracted and classified into some predefined classes. The CNN layers used in the network architecture are briefly explained below.

## 2.2.1 Convolutional layer

This layer acts as a feature extractor responsible for extracting the local features of the given image, video, speech or text. The layers have certain parameters whose task is to use the learnable kernels which are small in dimensionality. Despite being small in dimension these kernels are entirely spread on the depth of the input. The convolution layer uses the concept of parameter sharing where all neurons are allowed to recognize the exact same pattern by restricting them to share one single set [2]. The convolutional layer has set of multiple filters or the operations for recognizing the multiple features in a single layer; here each filter is a group of neurons which recognizes the patterns at the different location of the image or video.

In CNN the input data is fed to a convolutional layer which produces a two-dimensional activation feature map (Fig. 1). This feature map is the result of convolving each filter across the dimension of the input. The convolving operation is the scalar product for each value in the filter. Convolutional layers can also reduce the complexity of the CNN model by optimizing its output. For this, it uses few hyper-parameters like the depth, the stride and setting zero padding [2]. Stride decides how many convolution operations are conducted over the input. Padding hyper-parameter adds an additional 0 value to the border of the input.

## 2.2.2 Pooling layer

As convolutional layer detects the local feature map from the input data, the pooling layer reduces the dimensionality of each feature map [72]. The high dimensional features incorporate a large number of parameters while performing the training process. The pooling layer reduces the computational cost by reducing the number of parameters when the images are too large, however, at the same time it retains the important information. It generalizes the results from the convolutional filter making the detection of features invariant to scale or orientation changes. The pooling layer is not parameterized or in other words, the neurons in the pooling layer do not have weights or biases. Pooling layer comes with several variants like average pooling, max pooling, and sum pooling. Max pooling is a sample based discretization process that takes the largest element from the feature map. The pooling is performed over each feature map and the max function is used to scale the dimensionality of the feature. It basically reduces the size of the data because a large number of parameters involves large chunks of data. Thus, a smaller number of parameters are needed such that a CNN can still identify the image.

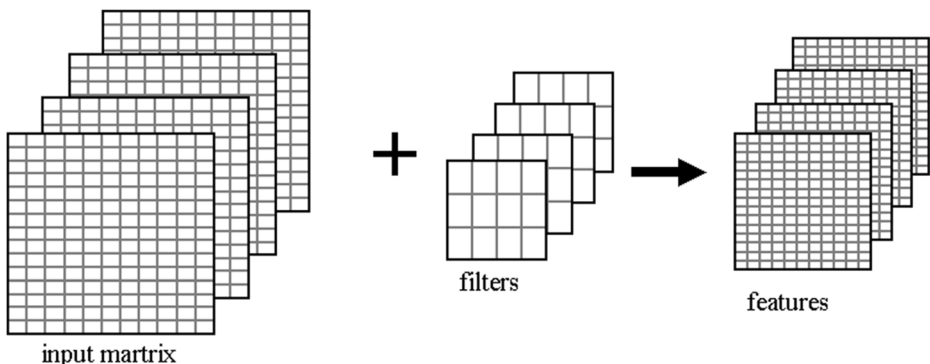
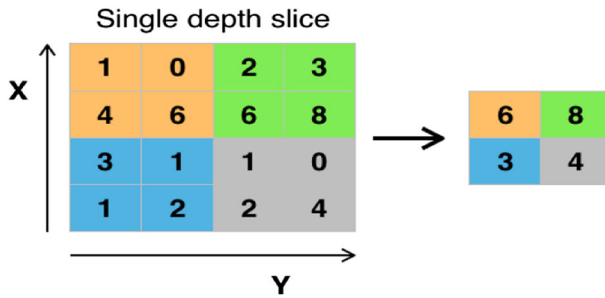


Fig. 1 Convolution operation





**Fig. 2** Max Pooling Operation

Figure 2 represents the max pooling operation performed on input data. Let's say there is a  $4 \times 4$  matrix representing the initial input and a  $2 \times 2$  filter that will convolve over the input. The stride value of  $2 \times 2$  is been selected which controls how the filter convolves around the input volume. For each of the regions represented by the filter, the max of that region is taken and a new, output matrix where each element is the max of a region in the original input is created.

### 2.2.3 Fully connected layer

The features extracted from the previous layers need to be classified into predefined classes, which is carried by fully connected layers. The fully connected layer is the last layer of the CNN architecture hence no other layers can be placed after this layer. The neurons have a full connection with the activations of the previous layer. The convolution layer and pooling layer extracts features and also reduces the number of parameters from the data, therefore fully connected layer is applied to generate an output which equals the categories or classes. A non-linear activation function called softmax is applied to the fully connected layer which outputs the probabilities distribution over class labels [25].

### 2.2.4 Dropout

Due to the high capacity and with limited training data the CNN models are very much prone to overfitting. Various regularization techniques are applied to prevent the over fitting such weight decay and augmentation of the input data. The other techniques are to stop the training as soon as the performance gets worsen or including the various kinds of weight penalties such as L1 and L2 regularization [64]. Hinton et al. [31] proposed dropout, another regularization approach that randomly sets a few activations within a layer to zero. The term dropout refers to dropping out units (hidden and visible) in a neural network with a certain probability.

## 2.3 CNN for emotion recognition tasks

The deep learning architectures for recognizing various facial expressions and emotions from the faces were reported in [2, 10, 12, 14, 17, 19, 22–25, 29–31, 33, 34, 38–40, 42–45, 51, 52, 56, 57, 59, 61, 63–65, 71–73, 75, 78, 81–84, 86, 87, 90, 91]. Liu & Li et al. [56] proposed a 3D CNN to detect specific facial expressions under the structured spatial constants. A 5 layer CNN for classifying emotions from the Cohn\_Kanade (CK+) dataset was presented by Lopes et al. [57]. Arushi and Vivek [70] used a VGG16 pertained network to classify images of human faces into discrete emotion categories. Hamster et.al. [27] proposed an architecture based on

multi-channel CNN for recognizing facial expressions on the JAFFE dataset where upper channel used convolutional filters, while Gabor like filters were used in the lower channels. Xie and Hu [88] proposed a feature redundancy reduced CNN to address the problem of facial expression recognition. For conducting the experiment, a facial expression database was collected and achieved it better results as compared to other expression recognition methods. Multiple deep CNNs were trained by Kim et al. [46] where they varied input normalization and network architectures. They also used the learning strategies ignorer to use the large external databases. G. Levi et al. [54] proposed a different CNN model for facial emotion recognition. They addressed the two problems in their work such as a small amount of data available for training deep CNNs appearance variation usually caused by variations in illumination. Local Binary Patterns (LBP) was used by them to transform the images into an illumination invariant. Samira et al. [15] used RNN combined with CNN architecture to predict emotion in the video called CNN-RNN architecture. A hybrid convolution-recurrent neural network was proposed in [41] for facial emotion recognition, where CNN extracts the relations within images and RNN considers the temporal dependencies existing in image during the classification. The CNN model was earlier applied to solve only the task of document classification and sentiment analysis. Till now only the coarse-grained features of text have been extracted from the text but a less amount of contribution has been paid to detect the fine-grained emotion features from the text. Keeping this fact in mind CNN based methodology has been proposed to deal with fine-grained emotion features of the text with the help of sequence based CNN architecture.

### 3 Data Collection & Annotation Methodology

Section 3.1 and 3.2 deals with the data collection methodology and manual annotation task of the corpus respectively. An Inter-annotator agreement (IAA) is performed over the annotated corpus in section 3.3.

#### 3.1 Building a Corpus for emotion detection

The objective of this research work is to automatically recognize emotions from multimedia textual content. The foremost aim is to enable CNN methodology for the emotion detection of the text. This calls for an appropriate large corpus of text that is rich in emotions apart from short texts like tweets, reviews, headlines etc. The emotion analysis of short texts is challenging because they contain limited contextual information in them. This type of text data is somewhat restricted to be applied in sentiment analysis task because only the coarse grained features need to be extracted. For example, whether the text has a positive or negative sentiment. However, to extract and detect the fine-grained features it requires a more extensive dataset. Therefore, to encounter this limitation of short text it is assumed that TV shows are a rich resource of natural expression of feeling, thoughts, and opinions. For the experiments conducted and reported in this paper the transcript of the famous TV show Charmed [<http://www.charmedtranscript.tv/>] is used as a source of emotional expressions.

There are total of eight seasons of the TV show from which three seasons are extracted for building the corpus. Each season consists of episodes, each episode contains scenes and each scene includes utterances. The corpus is called the charmed dataset throughout the paper. The corpus consists of 3 seasons, 66 episodes, 847 scenes, and 13,354 utterances. Consider a

corpus of  $U = \{u_1, u_2, \dots, u_n\}$ , utterances, each of which belong to  $E = \{e_1, e_2, \dots, e_n\}$  emotions. The task is to assign the label  $l$  for each sentence of  $K$  by appointing  $A = \{a_1, a_2, \dots, a_n\}$  annotators. Psychologists have argued on different emotions models [16, 35, 68, 69, 85]. For the task of annotation, emotion categories are borrowed from Ekman's emotion model and the corpus is manually annotated as happiness, sadness, surprise, anger, fear and disgust. If a text has no emotion, it is annotated as neutral. The emotions considered for this study are arranged into a tree structure, as shown in Fig. 3. The figure helped annotators to understand how emotions are related to each other. Table 2 describes a scene from a particular episode containing seven utterances and annotated emotion along with that.

### 3.2 Emotion annotation task

This section presents the data annotation task to annotate a corpus of 3 seasons of a charmed dataset. There is a lot to consider when choosing the right annotation. It is virtually impossible for all annotators to annotate every sentence exactly the same. It is good to have a reliable annotation by incorporating more than one annotator for each class of label otherwise it results in noisy annotations. So, to accomplish the goal total 3 annotators were selected who worked independently on the corpus for one and a half month. One of them was a researcher itself and the other two were the colleagues who are expert in the English language. All the annotator belongs to the age of 24 to 30 and shares a comparable educational background (university graduates or research scholars). Before beginning the annotation procedure, the corpus is divided into four batches, where each batch is assigned to one annotator every week. Table 3 describes the total number of utterances and scenes in each annotation batch.

For the annotation task, Label box[<https://app.labelbox.io/>], a free and open source online annotation tool was used. The transcript data retrieved from the web is split into speaker and utterances column. A generated comma separated file consisting of speaker and utterances is created and uploaded into the Label box. The emotion categories were assigned in the form of buttons presented in horizontal alignment in order anger, fear, disgust, happiness, sadness, surprise and none. Each utterance was presented one at a time to permit the annotators to work on the corpus. Annotators were instructed to provide an appropriate emotion label that best describes the emotion of a speaker's utterance. Annotators were not provided any training, but few samples of annotated sentences were shown to them before the annotation task. They followed written instructions describing how to perform the annotation with a definition of each label. To make the annotation process more convincing, annotators were asked to annotate the sentence on the basis of the feel of text and not to rely much on the keywords. As it is already discussed that the corpus used for annotation task is a TV transcript, therefore few TV show footages were also shown to the annotators. This was done because the annotators have to work on text data, therefore showing the footage of shows help them to

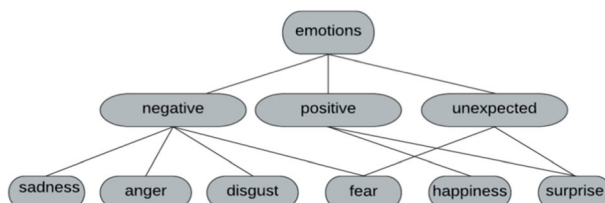


Fig. 3 Emotions Ontology

**Table 2** An example of the scene from a charmed transcript with emotion annotation

| Speaker  | Utterance                                                           | Annotator 1 | Annotator 2 | Annotator 3 | Annotator 4 |
|----------|---------------------------------------------------------------------|-------------|-------------|-------------|-------------|
| Brent    | Please be careful with these monsters.                              | Fear        | Sadness     | Fear        | Fear        |
| Phoebe   | Don't worry, I will.                                                | Neutral     | Fear        | Neutral     | Sadness     |
| Davi     | What a pleasure. I can't tell you<br>how long I've waited for this. | Happiness   | Happiness   | Neutral     | Sadness     |
| Grimlock | Shut up!                                                            | Angry       | Angry       | Disgust     | Angry       |
| Prue     | Wow, you told him.                                                  | Surprise    | Sadness     | Surprise    | Surprise    |
| Piper    | Oh my God, this could be very dangerous.                            | Fear        | Sadness     | Sadness     | Fear        |

make the appropriate judgment. To avoid any kind of annotation, bias the annotators were asked to work independently.

The annotation task shows that some of the sentences exhibit more than one emotion - consider the sentence in Fig. 4, which carries the emotions of both and fear and happiness. In this type of cases, the annotators were asked to identify the essence of the primary emotions presented in the sentence.

It was also found that some sentences could not be attributed to any of the basic categories (Fig. 5). In this case, the annotators were instructed to annotate this type of sentences as neutral.

### 3.2.1 Annotator's performances

Though annotators belong to the same educational background somehow, they differ in their opinions, thoughts and another perspective of their thinking while dealing with annotation task. Annotators were instructed to complete the task in one and a half month and a meeting were fixed with them after the completion of the period. The summary of the corpus was submitted by all three annotators in a tabular form. The distribution of emotion categories by all three annotators are shown in Table 4. In Fig. 6 precise distribution of annotations between three annotators was observed, about 45%–50% of the data yields neutral and surprise type of emotions, whereas the sentences depicting fear and anger, are less in number. This variation in emotion distribution can be observed because while annotating a sentence the annotators get confused between the feel of two emotions, i.e. *fear* was confused by the *surprise* emotion and also the *anger* got confused with the *disgust* emotion. The nature of the show is “magic” which is the reason for such type of imbalance observed in the emotion annotation task.

### 3.3 Inter annotation agreement

In this section, the task is to find the agreement between the different annotators using some metrics which quantifies these agreements. For this, inter annotation agreement is applied on the annotated dataset which is the measurement of the extent to which data annotators assign the same annotation or label to the same variable. There is a plethora of statistics that a

**Table 3** The scenes and utterances divided into batch

| Type       | Batch 1 | Batch 2 | Batch 3 | Batch 4 | Total  |
|------------|---------|---------|---------|---------|--------|
| Scenes     | 238     | 211     | 192     | 206     | 847    |
| Utterances | 2752    | 3754    | 3589    | 3259    | 13,354 |

You better turn your cell phone off too. Wouldn't want it to ring and scare it away. (They turn off their phones.)  
This is a nice spot. How'd you find this place? You and Piper's sister spend some time here?

**Fig. 4** An example of mixed emotions

researcher can use to measure agreement among the annotators, for example, percent agreement [3], Cohen's kappa [13], Fleiss's kappa [18] and Krippendorff's alpha [48]. Figure 7 presents the proportion value of emotion categories between the pair of annotators. It is clearly visible that most of the emotions depicted in the TV show are surprise and sadness, whereas annotators marked most of the sentences as neutral, because of the fact that some sentences do not give any feel of emotions.

Table 5 depicts the percent agreement between the different pairs of annotators. Next kappa and alpha coefficient are calculated for each pair of annotators using [<http://vassarstats.net/kappa.html>] and an online tool [21] respectively. The value of Cohen's kappa for each two pair of annotators is shown in Table 6. It can be observed that the value of  $\kappa$  varies across the annotators. This happens because of the fact that annotators have a different perspective of analyzing the sentences for emotion labeling. The average kappa value is 0.4101 which shows the moderate agreement. The commonly used kappa statistics scale [50] is presented in Table 7. Table 8 represents the alpha coefficient between all three pairs of annotators. The calculated Krippendorff's alpha coefficient is equal to 0.46 and it was reported moderate in [5]. The resulting Kappa and Alpha values are moderate but fine to carry out in further research. Full agreement and the quality of annotation can be achieved by incorporating more annotators.

### 3.4 Rule formation for final Corpus

The task of manual annotation is to provide training data for a classification model. The quality of training data, in turn, improves the accuracy of the model or the algorithms. The most straightforward strategy for assigning appropriate labels to the dataset was done by applying a voting scheme where the most assigned and popular label is assumed to be correct. At first, the dataset was partitioned into four parts using few rules (Table 9).

- It can be observed that in set 15,078 utterances were annotated accurately by the annotators; this means the agreement between the three annotators is 100%. Therefore, these utterances were taken as it is in the final corpus along with their labels.
- In set 2 the annotator 1 and annotator 2 shows full agreement and complete disagreement with annotator 3, therefore annotations coming from both annotators 1 & 2 win the votes and take part in final corpus creation.
- Also, in set 3, the annotator 1 and annotator 3 shows the full agreement and complete disagreement with the annotator 2, therefore annotations coming from both annotators 1 & 3 win the votes and take part in final corpus selection. In the next step, the magnitude relative error (MRE) is calculated which measures the difference between previous annotations and final annotations of each annotator. The annotations generated by the annotator with minimum MRE are chosen as final labels for set 4 and 5.

I actually do think I have something. Uh, yeah, just hang in there, Piper.

**Fig. 5** An example of no emotions

**Table 4** Distribution of emotions w.r.t. annotators

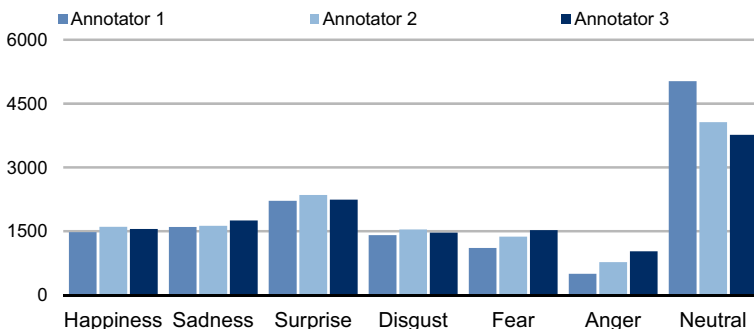
| Emotion Category | Annotation set 1 | Ratio (%) | Annotation set 2 | Ratio (%) | Annotation set 3 | Ratio (%) |
|------------------|------------------|-----------|------------------|-----------|------------------|-----------|
| Happiness        | 1481             | 11.085    | 1610             | 12.05     | 1557             | 11.654    |
| Sadness          | 1602             | 11.991    | 1629             | 12.193    | 1757             | 13.151    |
| Surprise         | 2216             | 16.586    | 2356             | 17.634    | 2244             | 16.796    |
| Disgust          | 1409             | 10.546    | 1543             | 11.549    | 1468             | 10.988    |
| Fear             | 1111             | 8.315     | 1375             | 10.291    | 1529             | 11.444    |
| Anger            | 505              | 3.779     | 777              | 5.815     | 1031             | 7.717     |
| Neutral          | 5030             | 37.649    | 4064             | 30.419    | 3768             | 28.203    |

### 3.5 Final Corpus analysis

Figure 8 shows the emotion distribution in the final charmed transcript corpus. Some of the sentences contain multiple emotions and the confusion of annotator to whether keep them into the emotion category or not was a key problem. Therefore, most of the sentences have been annotated as Neutral (33.3%). About 50% of dataset is yield together by neutral and surprise emotions. They are considered to be the most dominant emotions. The most common emotions were Surprise (17.1%), Sadness (15.7%) and Happiness (11.5%). The Anger (5.1%) emotion is relatively low as compared to Fear (7.7%) and Disgust (9.7%). The disgust often gets confused with sadness as the essence of disgust is disapproval of something which indirectly leads to sadness. The surprise emotion was found to be most typical as it sounds negative in a few utterances but positive many times. The more balanced emotion distribution is shown in Table 10, while considering the coarse-grained emotions, like positive, negative and neutral. The tree structure depicted in Fig. 3 is taken as reference. The ratio of negative emotions (38.1%) is high because of the fact that the show belongs to a magic genre.

## 4 Emotion detection methodology

The methodology of the proposed CNN framework assumes the existence of the annotated corpus based on respective emotions expressed. This methodology includes data collection, input data pre-processing, feature generation and emotion detection task. The pipeline of the proposed framework is organized in the following way (Fig. 9):

**Fig. 6** Precise distribution of emotions among three annotators

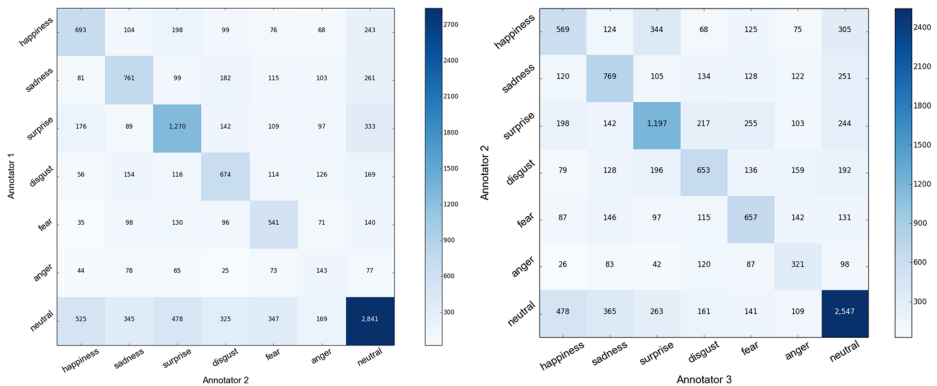


Fig. 7 Confusion matrix between Annotator 1 & 2 and Annotator 2 & 3

- **Corpus Building** This phase is the most important among the steps conducted in this study. A large corpus of sentences is extracted and manually annotated with the help of the annotators. The inter-annotator agreement is done to evaluate the annotated corpus which will be used for training the emotion detection model.
- **Pre-processing & Word Embedding** Before training the corpus into the model, the dataset is preprocessed by applying the NLTK module: tokenization, lemmatization, stop word removal and lower casing the sentences. Word embeddings are the numerical representation of contextual similarities between words. In this phase, the word embedding's using pre-trained word2vec pre-trained model have been created [58].
- **Emotion Detector** This phase is the core of the model, which trains the input data and classifies them into desired emotion category. Here a sequence based CNN has been used where the extracted feature maps from training data are passed in a sequential manner. The advantage of the proposed model is that the sequence information of the sentences is leveraged to improve the classification task.

#### 4.1 Dataset pre-processing

Before training the data into the model, the data is preprocessed in a way to discard the information which is irrelevant or futile for the feature maps. Pre-processing reduces the size of training dataset which in turn speeds up the training process. The importance of text processing on the performance of the proposed CNN model has been evaluated. Pre-processing involves a series of techniques which should improve the next phases of elaboration, in order to achieve better performances. The text data cannot be fed as raw

Table 5 Percent agreement between the pairs of annotators

| Annotator pairs           | Percent Agreement |
|---------------------------|-------------------|
| Annotator1 v/s Annotator2 | 0.518             |
| Annotator2 v/s Annotator3 | 0.502             |
| Annotator1 v/s Annotator3 | 0.544             |
| Average                   | <b>0.508</b>      |

**Table 6** Inter-Annotator Agreement between annotators

| Annotator pairs | Observed Agreement | Agreement by chance | Kappa         |
|-----------------|--------------------|---------------------|---------------|
| A1 v/s A2       | 0.5184             | 0.1949              | 0.4019        |
| A2 v/s A3       | 0.5027             | 0.1746              | 0.3975        |
| A1 v/s A3       | 0.5443             | 0.1985              | 0.4314        |
| Average         | <b>0.5218</b>      | <b>0.1894</b>       | <b>0.4101</b> |

input directly to the CNN Model as it contains noisy data. This raises the dimensionality of the problem and makes the classification process more difficult.

The basis constituent of texts is words and there is a need to transform the text sequences into the word. So as a first step the utterances (sentences) are pre-processed. At first, a tokenizer is used which is an NLP module to tokenize the text into words. The aim is to reduce the complexity of data, detecting and removing irrelevant and noisy elements from the data. It splits the longer strings of texts into smaller parts or tokens. In the next step, the process of lemmatizing is performed over the tokens which replace a given token with its corresponding lemma. Finally, all the text is converted to lower case, and extra blank spaces digits, exclamation marks, and question marks are removed.

Every one of the tasks in this module is executed to endeavor to make the text data uniform. This is important in light of the fact because features are chosen only when they exceed a certain frequency in the data set during the classification. In this way, after the essential preprocessing task, having distinctive words written in the same way helps them to achieve better classification. After the dataset have been pre-processed, the resulting sentences (utterances) become the instances of a new training set. Then, such a dataset is used for training the proposed model.

## 4.2 Sentence matrix and word embedding

Given the dataset,  $D$  consists of a set of utterances  $u = (u_1, u_2, \dots, u_n)$ . Each of the utterances is handled as a series of  $n$  words  $w = [w_1, w_2, \dots, w_n]$ . Each word of an utterance is associated with a vector representation. Let  $V$  be vocabulary and words are represented by a vector  $v \in \mathbb{R}^{1 \times d}$  looked up into  $V^{em}$  where  $em$  is the embedding size. The word embedding matrix  $X \in \mathbb{R}^{d \times V}$  is formed by concatenating embeddings of all words in  $V^{em}$ . The matrix  $X$  is passed as an input in the proposed emotion detection model. In this experiment, the word2vec model is used for constructing vector representations of words. The words in the resulting vocabulary are represented by  $d = 300$ -dimension vector.

**Table 7** The K value can be interpreted as follows

| Kappa     | Agreement        |
|-----------|------------------|
| <0        | Less than chance |
| 0.01–0.20 | Slight           |
| 0.21–0.40 | Fair             |
| 0.41–0.60 | Moderate         |
| 0.61–0.80 | Substantial      |
| 0.81–0.99 | Almost perfect   |



**Table 8** Krippendorff's Alpha between all annotators

| # of Annotators | Annotators Pair                             | Alpha |
|-----------------|---------------------------------------------|-------|
| 3               | Annotator 1 v/s Annotator 2 v/s Annotator 3 | 0.46  |

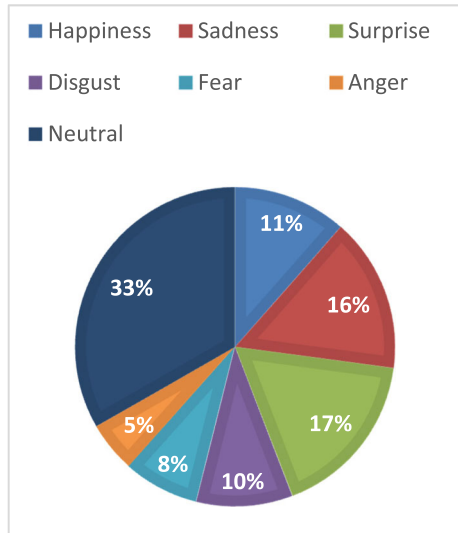
### 4.3 Proposed emotion detector model

This section introduces the main steps to be performed for carrying out the emotion detection. The emotion detection model is shown in Fig. 10. The feature extraction using the CNN model with attention mechanism for training the corpus is presented in the section. The CNN architecture is induced from the model presented by Kim et al. [45]. The corpus contains a sequence of utterances which allows solving the sequence classification task. In this section, a sequence based CNN model has been introduced which utilizes the previous sentence in order to detect emotion from the current sentence. Multiple feature extraction channels have been used in the proposed model for emotion detection task. Given an utterance  $u$ , the proposed model computes a score  $s$  for each seven emotion labels  $l \in L$ . The model takes the sequence of utterances as input and processed input is passed through a set of layers. The set of layers extracts the features from the sequence of words. The set of words of the utterances are converted into high dimensional vectors using pre-trained word embeddings. In CNN, filters (kernels) plays a key role as feature extractors such as semantic and context features from the sequence of words.

The proposed model is composed of one-dimensional convolution layer including nonlinear activation function, the max pooling operations to extract the feature from the input data and a fully connected layer with a softmax classifier to classify them into six emotion classes. The features are extracted by convolving the filters over the sequence of words. The convolution layer extracts the local and context features from the vector representation of the words and passes it to the fully connected layer via max pooling layer. The CNN context features are then fed as output to the attention model, which automatically focuses on the words that have a higher effect on the classification and captures the most important semantic information in a sentence. After widely employed in the field of computer vision [60], the attention mechanism has become popular in natural language processing. It has recently demonstrated success in a wide range of tasks such as machine translation [6], question answering system [28], speech recognition, etc. [11]. In the field of natural language processing [89, 92] have achieved the state of art performance for classifying multiple sentences. The attention mechanism model is adapted from the work conducted by [26] for text classification using CNN and Recurrent Neural Network. The proposed model consists of 5 CNN layers, such as two one-dimensional

**Table 9** Rules for the voting scheme with their annotation counts  $a_1$ = annotator1,  $a_2$ = annotator 2 and  $a_3$  = annotator 3

| Set | Rules                                                     | No. of utterances | Ratio in (%) |
|-----|-----------------------------------------------------------|-------------------|--------------|
| 1   | $a_1 = a_2 = a_3$                                         | 5078              | 38.03        |
| 2   | $(a_1 = a_2) \wedge (a_1 \neq a_3) \wedge (a_2 \neq a_3)$ | 2457              | 18.4         |
| 3   | $(a_1 = a_3) \wedge (a_1 \neq a_2) \wedge (a_3 \neq a_2)$ | 2825              | 21.15        |
| 4   | $(a_2 = a_3) \wedge (a_2 \neq a_1) \wedge (a_3 \neq a_1)$ | 1867              | 13.98        |
| 5   | $a_1 \neq a_2 \neq a_3$                                   | 1127              | 8.44         |

**Fig. 8** Distribution of emotions in final corpus

convolutional layers, two pooling layers, and one fully connected layer followed by an attention mechanism layer.

**Convolution layer** This is the core building block in the proposed CNN model which extracts the features from the input data by applying the convolution filter operations. The one-dimensional convolution layer has been applied to deal with the text data. The set of  $f$  filters,  $F \in R^{d \times w}$  filter weights have been applied to a sliding window of length  $w$  over each sentence. These filters are convolved with the  $n$  words to obtain the feature maps. The numbers in the filter are the neuron's weight (parameters) which are updated at each training session, however, initially, the filters are initialized randomly. The new matrix  $c \in R^{em-w+1}$  feature vector is generated after applying a convolution operation between input  $X \in R^{d \times em}$  and a  $F \in R^{d \times w}$  which are passed forward in the network:  $c_j = (X * F)_j$ . The feature vector  $c$  is aggregated over all the  $f$  no. of filters of form a feature map matrix  $C \in R^{f \times (em-w+1)}$ . In addition to this, a  $b$  bias vector is also applied in the convolution result for each feature map  $c_j$ . This is done to help the network learn an appropriate threshold. The filter size of the two convolutional layers is chosen to be 5.

**Table 10** Coarse-grained and Fine-grained emotion distribution

| Coarse-Grained Emotions | Fine-Grained Emotions | Ratio (in %) |
|-------------------------|-----------------------|--------------|
| Positive                | Happiness             | 28.6         |
| Negative                | Surprise              | 38.1         |
|                         | Sadness               |              |
|                         | Disgust               |              |
|                         | Fear                  |              |
|                         | Anger                 |              |
| Neutral                 | Neutral               | 33.3         |

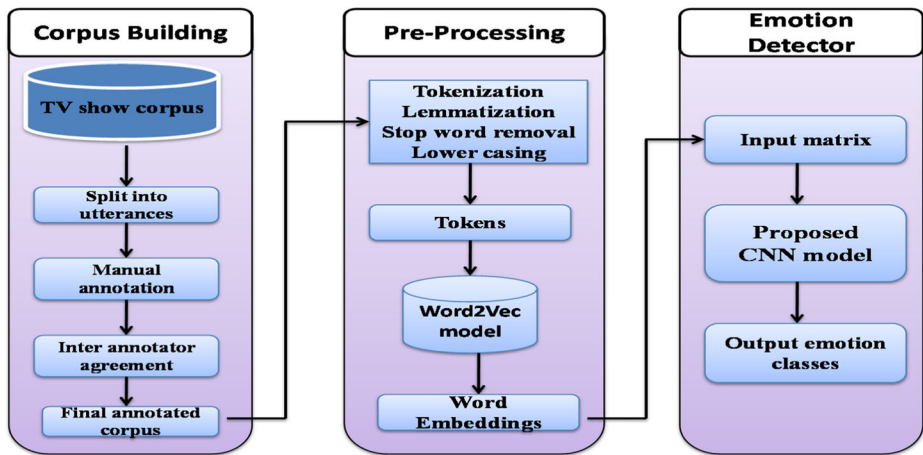


Fig. 9 Proposed Framework for Emotion Detection

**An activation function** Among the list of well-known activation functions like sigmoid function, hyperbolic tangent(tanh) and rectified linear unit(ReLU), a Rectified Linear Unit(ReLU) have been applied since it increases the accuracy of the results as well as speed up the training [62]. The non-linear activation function allows a model to handle non-linear

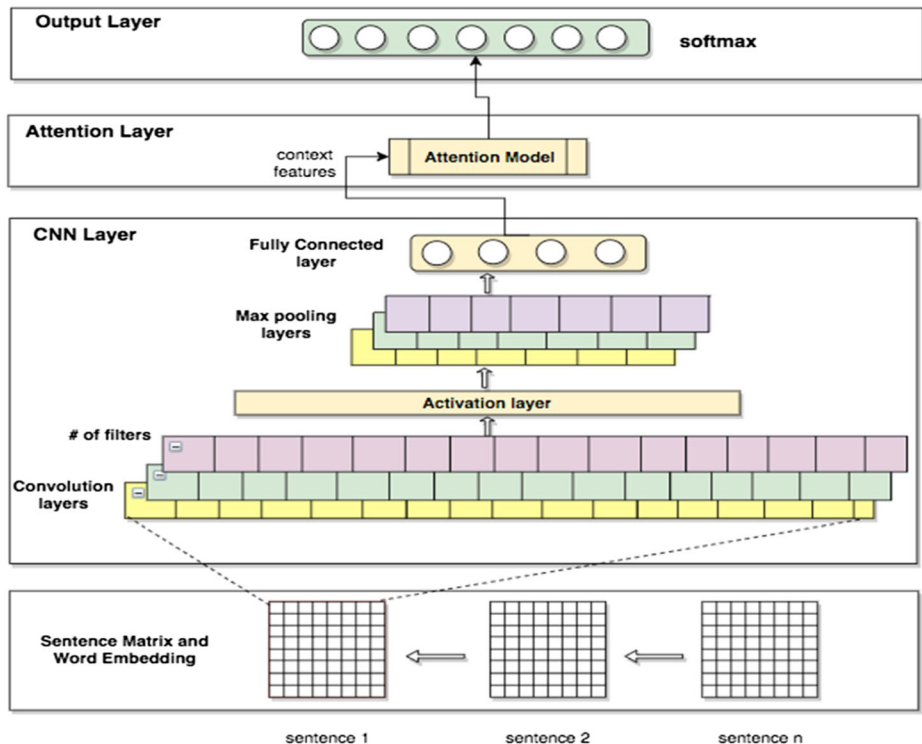


Fig. 10 Proposed CNN Model for Emotion Detection

data which cannot be separated by a single line after they have been classified. ReLU is defined as:  $f(x) = \max(0, x)$ .

**Max pooling** After applying the ReLU activation function to the convolutional layers, the output is fed into the pooling layer. A max pooling operation is applied to the result of convolution layers which returns the maximum value. The resulting pooled feature map matrix  $P$  is  $P = [p(\alpha(c_1)), p(\alpha(c_2)), \dots, p(\alpha(c_n))]$ , where  $p$  is a pooling operation,  $\alpha$  is the ReLU operation and  $c_i$  where  $i = (1, 2, \dots, n)$  is the feature map. In the proposed model pooling size of value 2 has been considered. As a result, the feature map is passed to a fully connected layer to reshape the output of the CNN layer.

**Attention model** An attention model shown in Fig. 11 takes the matrix  $Y$  consisting of the output vectors ( $y_1, y_2, \dots, y_T$ ) that the fully connected layer generates, where  $T$  is the length of the sentence and a context  $c$ . It returns a vector  $z$  which is supposed to be the summary of the  $y_i$  focusing on the information linked to the context  $c$ . Simply it returns a weighted arithmetic mean of the  $y_i$  and the weights are chosen according to the relevance of each  $y_i$  given the context  $c$ .

Few steps were taken into consideration:-

- 1) First, the input  $c$  as the context and the  $y_i$  the feature vectors are recognized.
- 2) In the next step,  $u_1, u_2, \dots, u_n$  is computed with a tanh layer. This is the computation of an aggregation of the values of  $y_i$  and  $c$ .

$$u_i = \tanh(W_{cu}c + W_{yu}y_i)$$

- 3) Then, weights using a softmax is computed which behaves almost like an argmax. Here  $a_i$  are the softmax of the  $u_i$  projected on a direction. So, the softmax can be thought of as the max of the relevance of the variables, according to the context.

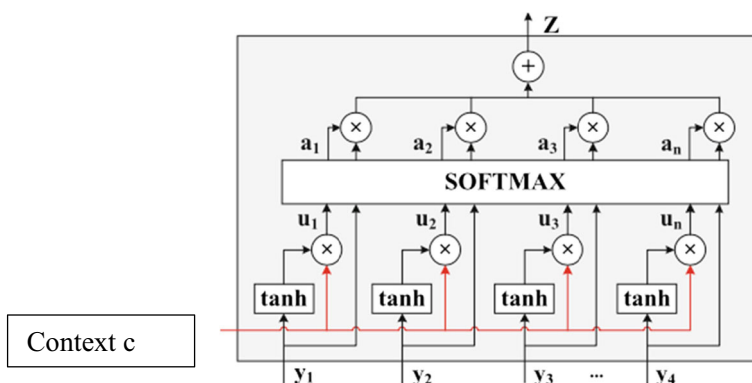


Fig. 11 Attention Model [26]

**Table 11** Hyper-parameter values for LSTM model and proposed CNN model

| Hyper-parameter  | Notation | # of values | Model              |
|------------------|----------|-------------|--------------------|
| number of epochs | ep       | 30,50,70    | Both models        |
| batch size       | b        | 100,128,256 | Both models        |
| hidden units     | hu       | 100,150,200 | Both models        |
| activation       | a        | relu        | Both models        |
| no. of filter    | nf       | 32,64,128   | Proposed CNN model |

$$a_i = \exp((w_u, u_i))$$

- 4) The output  $z$  is the weighted arithmetic mean of all the  $y_i$ 's, where the weight represents the relevance for each variable according to the context  $c$ .

$$z = \sum_i a_i y_i$$

The attention model chooses the variable the most correlated with the context.

**Output layer** The layer takes the output of the attention layer as input and computes  $N$ -dimensional vectors where  $N$  is the number of class labels is provided as outputs. A softmax classifier has been used to compute the probabilities to predict the output labels (or emotions).

$P(e) = \text{softmax}(X * W_j + b_j)$ , where  $P(e)$  is the probability distribution over labels,  $X$  is the input from the previous layers,  $W_j$  is the weight vector and  $b_j$  are the bias. As a cost function,

**Table 12** Results on altering hyper parameters for LSTM model

| Hyper-parameter             | EPOCHS = 30             |        | EPOCHS = 50             |        | EPOCHS = 70             |        |
|-----------------------------|-------------------------|--------|-------------------------|--------|-------------------------|--------|
|                             | Prediction Accuracy (%) | Loss   | Prediction Accuracy (%) | Loss   | Prediction Accuracy (%) | Loss   |
| b = 100, a = relu, hu = 100 | 58.56                   | 1.579  | 64.12                   | 0.9542 | 67.13                   | 0.8292 |
| b = 100, a = relu, hu = 150 | 60.67                   | 1.445  | 65.64                   | 0.9390 | 67.42                   | 0.7634 |
| b = 100, a = relu, hu = 200 | 60.52                   | 1.222  | 66.72                   | 0.8752 | 68.77                   | 0.7101 |
| b = 128, a = relu, hu = 100 | 62.63                   | 1.142  | 64.97                   | 0.8474 | 68.33                   | 0.6802 |
| b = 128, a = relu, hu = 150 | 63.72                   | 0.9901 | 65.80                   | 0.8126 | 68.16                   | 0.6429 |
| b = 128, a = relu, hu = 200 | 64.12                   | 0.9393 | 66.45                   | 0.7633 | 69.24                   | 0.5982 |
| b = 256, a = relu, hu = 100 | 64.81                   | 0.9124 | 65.36                   | 0.7255 | 70.77                   | 0.5714 |
| b = 256, a = relu, hu = 150 | 65.45                   | 0.8665 | 68.77                   | 0.7212 | 72.15                   | 0.5527 |
| b = 256, a = relu, hu = 200 | 64.67                   | 0.8201 | 70.53                   | 0.7123 | 71.42                   | 0.5412 |

**Table 13** Results on altering hyper parameters for proposed CNN model

| Hyper-parameter             | EPOCHS = 30            |        | EPOCHS = 50            |        | EPOCHS = 70            |        |
|-----------------------------|------------------------|--------|------------------------|--------|------------------------|--------|
|                             | Prediction Accuracy(%) | Loss   | Prediction Accuracy(%) | Loss   | Prediction Accuracy(%) | Loss   |
| b = 100, a = relu, nf = 32  | 60.23                  | 0.8427 | 62.28                  | 0.6556 | 69.56                  | 0.5542 |
| b = 100, a = relu, nf = 64  | 61.34                  | 0.8353 | 66.25                  | 0.5165 | 72.24                  | 0.5323 |
| b = 100, a = relu, nf = 128 | 64.13                  | 0.8224 | 69.57                  | 0.5664 | 74.45                  | 0.5255 |
| b = 128, a = relu, nf = 32  | 61.66                  | 0.7644 | 67.45                  | 0.5056 | 71.12                  | 0.5112 |
| b = 128, a = relu, nf = 64  | 63.07                  | 0.7588 | 71.34                  | 0.5932 | 74.04                  | 0.4967 |
| b = 128, a = relu, nf = 128 | 66.28                  | 0.7318 | 72.50                  | 0.5341 | 76.67                  | 0.4621 |
| b = 256, a = relu, nf = 32  | 62.98                  | 0.7156 | 69.21                  | 0.5235 | 72.65                  | 0.4432 |
| b = 256, a = relu, nf = 64  | 68.67                  | 0.6873 | 73.32                  | 0.5464 | 78.67                  | 0.4289 |
| b = 256, a = relu, nf = 128 | 69.73                  | 0.5856 | 75.54                  | 0.5212 | 80.41                  | 0.4285 |

categorical cross-entropy loss function has been used because of its capability to deal with multi-class classification. To alleviate the overfitting dropout with probability 0.5 is applied after each convolutional layer and the last layer.

## 5 Training process and evaluation

This section is devoted to the training mechanism and experimental evaluation of the proposed classification model in order to generalize the training dataset. The pre-processed dataset is used for training the proposed model and in order to justify the dataset, it was also trained over baseline random forest classifier (RF) and LSTM [32] model. Finally, the evaluation results obtained from different classification modules are compared with each other, in order to evaluate the efficiency and effectiveness of each technique.

The preliminary experiment on the estimate of the ability of the proposed model to generalize to newly collected data has been conducted. The dataset consists of 13,354 data rows which consist of a single label assigned to them. For evaluating the proposed model on

**Table 14** The performance of different models (in %)(Training Accuracy, Precision, Recall, F1-score)

| Models                                                   | Training Accuracy | Precision | Recall | f1-score |
|----------------------------------------------------------|-------------------|-----------|--------|----------|
| Random Forest(RF)                                        | 51.22             | 52        | 52     | 52       |
| LSTM                                                     | 72.15             | 71.31     | 70.21  | 70.76    |
| Proposed CNN model (fine-grained emotions(7 emotions))   | 80.41             | 80.34     | 0.8017 | 80.25    |
| Proposed CNN model (coarse-grained emotions(3 emotions)) | 83.32             | 83.12     | 0.8278 | 82.94    |

**Table 15** The performance of different models (in %)(Testing Accuracy)

| Models                                                    | Accuracy |
|-----------------------------------------------------------|----------|
| Random Forest(RF)                                         | 45.22    |
| LSTM                                                      | 59.56    |
| Proposed CNN model (fine-grained emotions(7 emotions))    | 77.54    |
| Proposed CNN model (coarse grained emotions (3 emotions)) | 80.99    |

different parameters, the dataset has been divided into three sets. The sklearn library of python language has been used to split the dataset. As the dataset is small therefore, training and the testing dataset is divided into 90% and 10% respectively. The manual tuning of hyper-parameter has been performed in the experiments therefore to get the best parameter for the proposed model the validation set (20%) was derived from the remaining 90% training set.

The same amount of training and testing data is applied to the LSTM model except the random forest classifier. For random forest classifier, the words are represented using word2vec representation and trained in the classifier with 100 estimators. The model achieves the training accuracy of 51.22% with precision, recall, f1-score 0.52, 0.52, 0.52 and test accuracy of 45.22% summarized in Table 14 and Table 15 respectively.

Deep learning performance is highly dependent upon the choice of hyper-parameters such as the number of layers, filter size, hidden units, dropout rate, weight parameters, learning rate, etc. Next, the effects of the several hyper-parameters are evaluated for the proposed model and baseline LSTM model. The sliding window  $w$ 's length is set to 5 and striding to 2. For preventing the model from overfitting a dropout rate of 0.5 is applied. In order to optimize the training mechanism, Adam Optimizer [47] optimization algorithm is used in this model which is an extension to stochastic gradient descent.

A categorical-cross entropy loss function is used in the proposed model to deal with seven emotion classes which measure the performance of a classification model. In section 5.1 testing is performed on the several hyper-parameters and experimental results have been discussed in section 5.2.

The baseline model and the deep learning models have been implemented using Keras,<sup>1</sup> an open-source neural network library written in Python. It is designed to enable fast experimentation with deep neural networks, it focuses on being user-friendly, modular, and extensible. Keras is run on top of TensorFlow<sup>2</sup> library which is an end-to-end open source platform for machine learning and deep learning. The Atom<sup>3</sup> editor is used for writing and executing the codes on the single MacBook air Sierra with 8 GB 1600 MHz DDR3 RAM and 1.8 GHz intel core i5 processor.

## 5.1 Tests for hyper-parameters

For the LSTM model each sentence is mapped into a vector domain called Glove embedding [66]. This technique encodes the words in real-valued vectors in a high dimensional space. Each word is mapped into 300 length vectors. Later the embedded words are fed into the LSTM layer followed by ReLU activation function and a dropout layer. Neurons in the last

<sup>1</sup> <https://keras.io/>

<sup>2</sup> <https://www.tensorflow.org/>

<sup>3</sup> <https://atom.io/>

**Table 16** Fine-grained test emotion classification for test dataset based on proposed CNN model

| Emotions  | Precision | Recall | f1-score      |
|-----------|-----------|--------|---------------|
| anger     | 0.77      | 0.65   | 0.70          |
| disgust   | 0.74      | 0.73   | 0.73          |
| fear      | 0.77      | 0.80   | 0.78          |
| happiness | 0.71      | 0.80   | 0.75          |
| neutral   | 0.61      | 0.66   | 0.63          |
| sadness   | 0.69      | 0.78   | 0.73          |
| surprise  | 0.75      | 0.63   | 0.68          |
| avg/total | 0.7243    | 0.7254 | <b>0.7248</b> |

dense layer have been kept seven as there are seven emotion categories along with activation function as “*softmax*” and optimizer as “*Adam*”. The LSTM layer contains memory units which are varied throughout the training along with other hyper-parameters.

Later on, the proposed model with word2vec word embeddings and attention layer to capture more semantic and context features from text dataset has been implemented. The functionality of the sequence based CNN is used to capture the features and learn the long dependencies at the same time. The input matrix is padded with all zeroes and length of the sentences are kept the same as the maximum length of the sentence. The filter sizes are varied while convolving over the words. Also, the number of filters and hidden units are varied for evaluating the effect of the dataset. The effect of activation function (ReLU) in the intermediate layers is also evaluated. The proposed model is trained for *ep* epochs and model’s performances are calculated with accuracy, loss, precision, recall and f1 measures. The following parameters have been experimented on the proposed CNN model and baseline LSTM, which are summarized in Table 11.

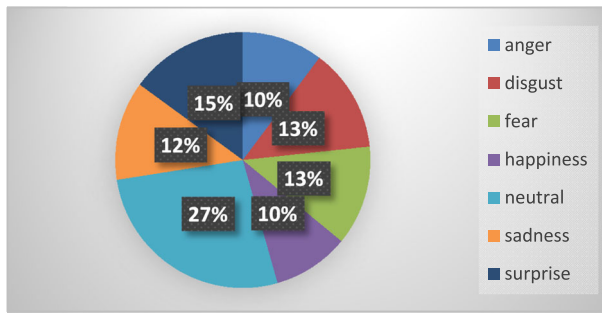
## 6 Results and discussion

In this section, the collected dataset’s performance has been evaluated on the proposed CNN model and LSTM model, and the performance of the models has been analyzed accordingly. The special effects of the hyper-parameters are evaluated in the experiments. As it is known that the dataset is imbalanced and where class “neutral” has the high number of data rows than other classes (Fig. 8), therefore the special care needs to be taken to deal with it. With the imbalance dataset, the training model may bias towards the classes which have a high number of instances. They tend to only predict the majority class data and minority classes are treated as noise and often ignored. Due to this reason some mechanism needs to be adopted to deal with this problem, at first the performance metrics like precision, recall and f1-score have been used. Another mechanism is to resample the dataset by performing the under sampling or over

**Table 17** Coarse-grained test emotion classification for test dataset based on proposed CNN model

| Emotions  | Precision | Recall | f1-score |
|-----------|-----------|--------|----------|
| Positive  | 0.76      | 0.73   | 0.70     |
| Negative  | 0.84      | 0.81   | 0.73     |
| Neutral   | 0.65      | 0.67   | 0.65     |
| avg/total | 0.75      | 0.73   | 0.74     |



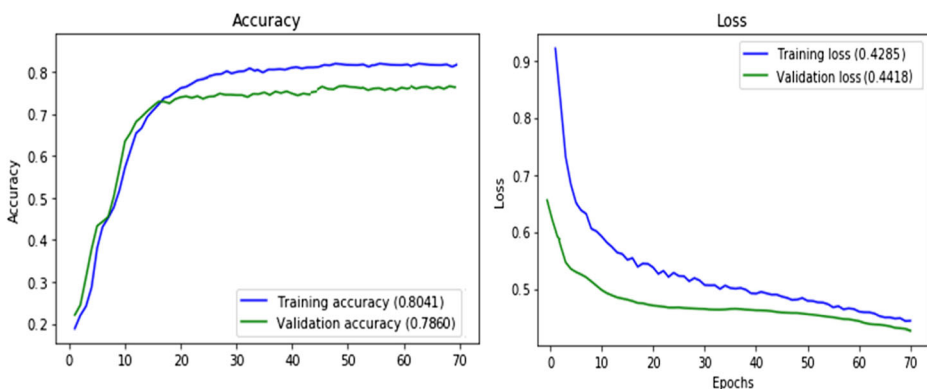


**Fig. 12** Distribution of emotions after SMOTE

sampling the dataset. Under sampling consists of deleting instances of the over-represented classes which leads to poor results if the dataset is less whereas oversampling adds copies of instances of the under-represented classes which leads to overfitting the model. Therefore, SMOTE (Synthetic Minority Over Sampling Technique) proposed by [9] has been applied. This technique identifies more specific regions in the feature space for the minority classes. For each minority class observation SMOTE calculates the K nearest neighbour representation depending upon the amount of oversampling needed. This technique is followed to avoid overfitting which occurs when exact replicas of minority instances are added to the main dataset. Figure 11 shows the distribution of emotions after performing the SMOTE mechanism. The sampling strategy in SMOTE function has been set to “minority” because the task is to oversample the under-represented classes.

Table 12 presents the prediction accuracy and loss of the LSTM model after training it on the charmed dataset. The effect of hyper-parameters is evaluated for epochs 30, 50 and 70 in the results of the LSTM model. It can be seen that when the number of hidden layers is increased from 100 to 150 at epoch 30 and 70 the prediction accuracy is found to be best among other hyper-parameter tunings. The model’s performance in the other case (i.e. hidden layers = 200) results in the decrease of performance. This happens because the model overfits to the training set i.e. it starts learning the training data but fails to generalize to new unseen data.

The prediction accuracy and loss of the proposed CNN model implemented for the prediction of emotions in the dataset is presented in Table 13. The effect of hyper-



**Fig. 13** Training accuracy and Loss for proposed CNN model

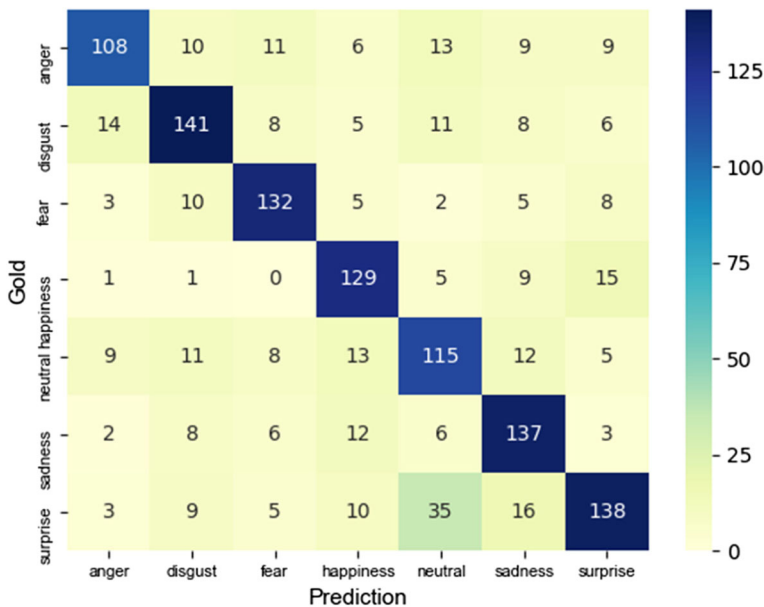


Fig. 14 Confusion matrix of the proposed model on the test set using the best hyper-parameters

parameters can be observed in the results as it helps to increase the prediction accuracy as well as reduces the loss value. The representational power of the model increases with an increase in a number of filters and in turn, improves the accuracy of the model. After the experimentation, the best results obtained when 128 numbers of filters and batch size for 256 were used in the model.

It can be observed that the increase in batch size improves the prediction accuracy. The reason behind the increase in accuracy is that the model learns by gradient descent an error function in the weight space which is parameterized by training examples. The small batch size makes the gradients more unstable and reduces the learning rate. With small batch size, the gradients are only a very rough approximation of the true gradients and hence take a long time to find a good solution in fewer training epochs. While larger batch sizes offer more parallelism and hence computational efficiency increases. Large batch size prevents the overfitting of data because of less redundancy in training data. Algorithmically speaking, using large batches in stochastic gradient descent allows the model to reduce the variance of the stochastic gradient updates by taking the average of the gradients in the batch, and this, in turn, allows it to take bigger step-sizes, which means the optimization algorithm will make progress faster and hence training accuracy increases. One hypothesis for why this happens is that with more labels, each batch must be large to be reliably representative of the dataset.

In deep learning, there might be an over-fitted model if trained so much on the limited training data. It means that the model does not learn the data, it memorizes the data. While training the

**Text:** Go away I don t want anything to do with you be a witch anymore

Fig. 15 Identification and highlighting the weight given to sample text using attention mechanism

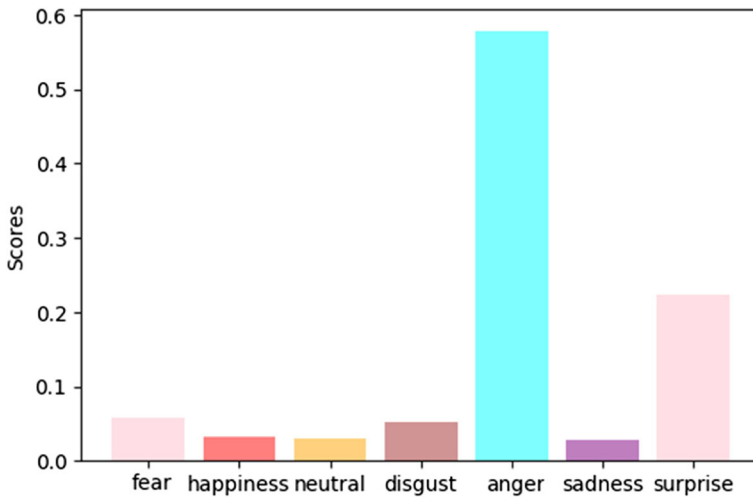


Fig. 16 Representation of emotion captured by attention mechanism

proposed model at first the epoch size is kept as 30 and the experiment is started by changing the batch size and number of filters. It was found that at batch size 256 and 128 number of filters the proposed model worked well whereas on the rest of the parameters it started underfitting the model. To overcome this problem, the epoch size is increased to 50 and found that the model is still working fine on 128 number of filters. To make the model more generalized to the unseen dataset epoch size is further increased to 70 where it is found that at batch size 256 and number of filters set to 128 the best value for accuracy (training and validation) and loss were found because after increasing the epoch size (more than 70) the proposed model started overfitting. This means the validation accuracy was constant on the increase in epochs. Therefore, the proposed model's training is early stopped at epoch size 70. However, to some extent overfitting of the model has been prevented by applying the drop out layer but due to the limitation of training data and increase in epochs, the model started overfitting.

Table 14 showcases the overall training performance of the baseline model, LSTM and proposed CNN model on the collected dataset. To compare the proposed model to other models' performances of LSTM and Random forest classifier are also been evaluated. The training accuracy is reported for the cases of 7 and 3 emotions (Table 10). It can be observed that the proposed CNN model outperformed LSTM and RF classifier and achieved an accuracy of 80.41% (for fine-grained emotions) and 83.32% (coarse-grained emotions) on a charmed dataset. The precision, recall, and f1-score for evaluating the performance of the proposed model and other states of the art models have also been analyzed.

Table 15 showcases the overall testing performance of the baseline model, LSTM and proposed CNN model on the evaluation set. The testing accuracy of the proposed model on fine-grained emotions is found to be 77.54% whereas the coarse-grained emotions are predicted with an accuracy of 80.99%.

Table 16 presents the performance results of the proposed model for each class of fine-grained emotions on the testing dataset. It can be observed that the precision of "anger" and "fear" is better as compared to other emotions after handling the class imbalance. Whereas the recall and f1-score of "happiness" are better than other emotion classes. Also, Table 17 presents the performance of proposed CNN model for each class of coarse-grained emotions

on testing dataset. Here the negative class has high precision and f1 score whereas the impact of majority class “*neutral*” have been decreased to some extent. This happens because of the fact that among 7 emotion classes 4 classes depicts negative emotions with 38.1% ratio (Table 10).

Prediction accuracy and loss of the proposed CNN model is shown in Fig. 12 over training v/s validation dataset. The charts illustrate the fine performance of the proposed CNN model where training accuracy and validation accuracy converges to near percentage values. The validation set used in the proposed model measure the ability of the model to generalize on unseen data. This claims that the hyper-parameter tuning strategy works fine to evaluate the model’s performance. From Fig. 12 it can also be seen that the training and validation loss of the proposed model is decreasing to a certain limit. However, more measures can be taken to decrease the loss by using regularization techniques and optimization techniques. The confusion matrix of proposed CNN model on the testing sets and the gold dataset is presented in Fig. 13. The results show that model can easily predict the true positive classes, however, few classes contradict with each other, “*anger*” is confused with “*disgust*” and “*happiness*” with “*surprise*”. This happens because the words occurring in anger and disgust have the same semantic meaning and as a result model confuses to label them. This can be handled by using more high dimension word embeddings and proper annotation technique.

Figure 14 shows the effectiveness of the attention mechanism model in selecting the informative words from the sample text used for testing the model performance. The sample text is randomly selected from the annotated dataset. Here the word “*go away*” is given more attention weight and highlighted in a green rectangle. The sample text is classified as the class “*anger*” however there arises a minor contradiction between “*anger*” and “*surprise*” class (Figs. 15 and 16).

## 7 Conclusions and future work

Emotion detection is a technology that quantifies mood, attitude, opinion, and emotion in digital multimedia, in images, video, audio, and text. Multimedia emotion detection can help businesses respond quickly and effectively to customer feedback and reviews. This work presents a framework based on the DNN for the task of emotion detection from the multimedia text data. A new corpus for emotion detection task was collected from a TV show transcript. The emotions were manually annotated with the help of other researchers and an inter-annotator agreement was performed to check the agreement of all. The collected dataset was trained on a proposed CNN model with attention mechanism which extracts the useful features from the text dataset. The proposed model is validated over the annotated dataset outperforms the baseline LSTM and Random forest classifier. The proposed emotion detection framework will help the business for learning the different sentiments and opinions of customers on certain products or review and social media monitoring as it allows us to gain an overview of the wider public opinion behind certain topics.

In the future, the work presented in this paper can be applied to the multi-label dataset where the intensity of each emotion class can be explored. Along with the text data the emoticons and text representing emoticons can also be considered for the task of emotion detection. The annotating task is most subjective task therefore in the future plan will be to assign more annotators to improve the inter-annotator agreement of the dataset. The hybrid

combination of CNN- LSTM can be implemented to fully evaluate the performance of emotion detection task, where LSTM can learn the long term dependencies from the text. In the present version of the manuscript focus has not been given on the aspect of time complexity of the training classifiers which will be considered in the future work. Also, the training dataset can be made larger by gathering more seasons of a TV show in order to avoid the over-fitting due to fewer data.

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